EC-350 AI and Decision Support Systems

Week 9
K-Nearest Neighbour Classifier

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Acknowledgements: Lecture slides material from Duda, Hart and Stork, Dr. Gavin Brown

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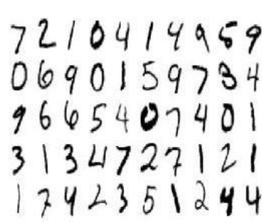
Problem Statement

- Why recognising rugby players is (almost) the same problem as ***recognising handwritten digits***



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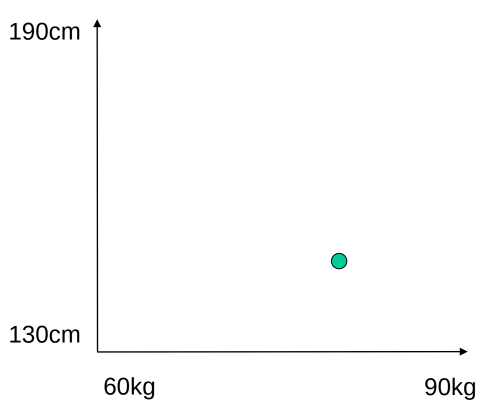
Problem Statement

Can we LEARN to recognise a rugby player and ballet dancer?

What are the “features” of a rugby player?

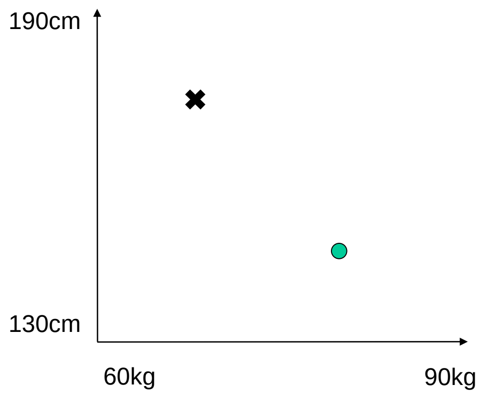
Features

Rugby players = short + heavy?



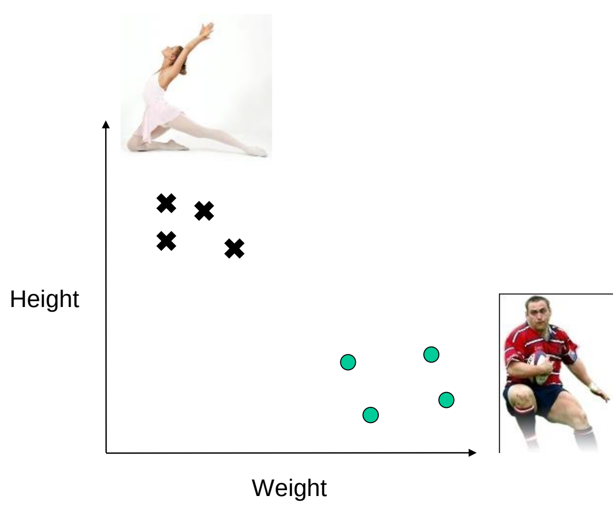
Features

Ballet dancers = tall + skinny?

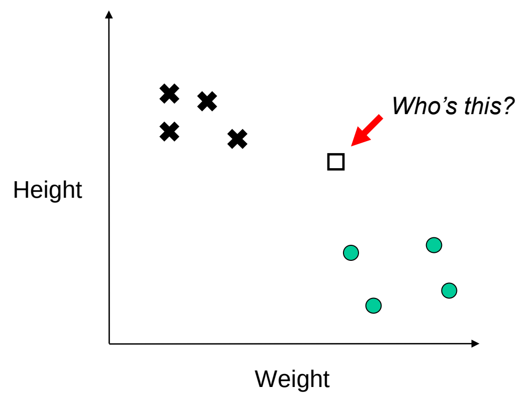


Feature Space

Rugby players “cluster” separately in the space.

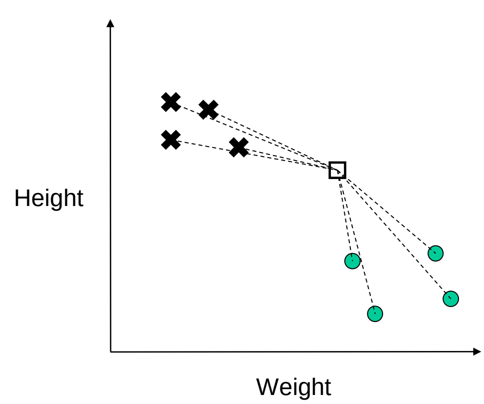


The K-Nearest Neighbour Algorithm



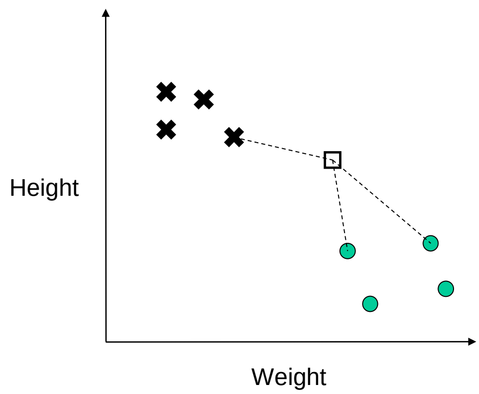
The K-Nearest Neighbour Algorithm

1. *Measure distance to all points*



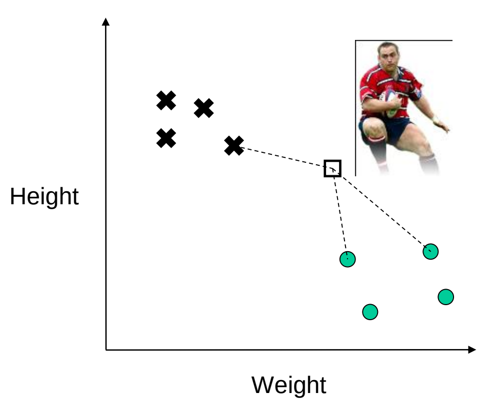
The K-Nearest Neighbour Algorithm

1. *Measure distance to all points*
2. *Find closest "k" points* ← **(here k=3, but it could be more)**



The K-Nearest Neighbour Algorithm

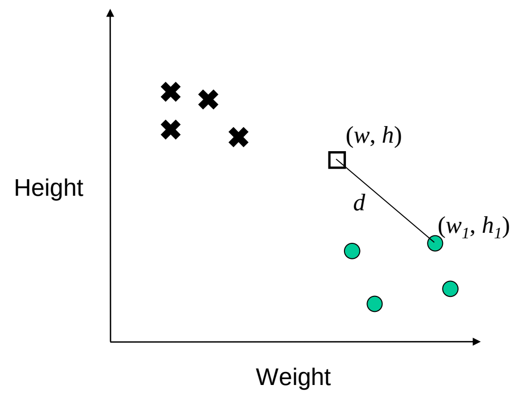
1. *Measure distance to all points*
2. *Find closest "k" points* ← **(here k=3, but it could be more)**
3. *Assign majority class*



Distance Measure

"Euclidean distance"

$$d = \sqrt{(w - w_1)^2 + (h - h_1)^2}$$



The K-Nearest Neighbour Algorithm

for each testing point

measure distance to every training point

find the k closest points

identify the most common class among those k

assign that class

end

- Advantage: Surprisingly good classifier!
- Disadvantage: Have to store the entire training set in memory

Distance Measure

Euclidean distance still works in 3-d, 4-d, 5-d, etc.…

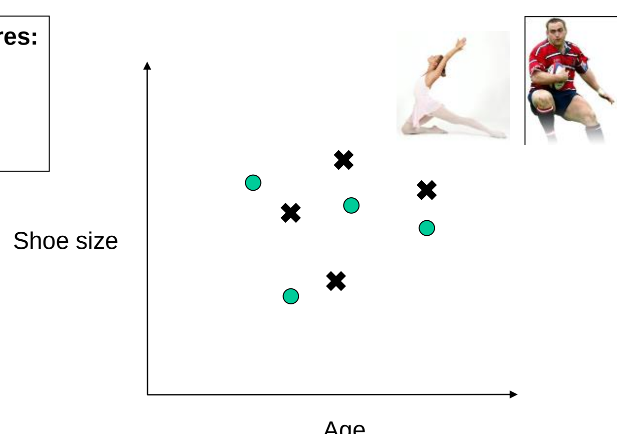
$$d = \sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2}$$

x = Height
y = Weight
z = Shoe size

Features

Choosing the wrong features makes it difficult
Too many features
It's computationally intensive

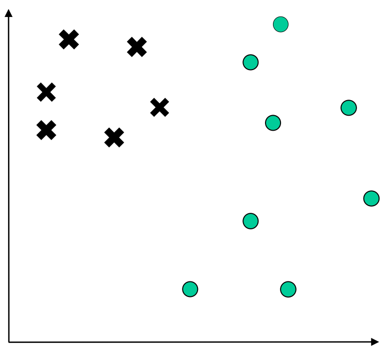
Possible features:
- Shoe size ✓
- Height
- Age ✓
- Weight



?

Over-fitting

An **Important** Concept in Machine Learning



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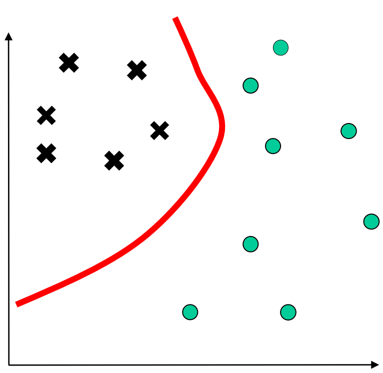
Over-fitting

An **Important** Concept in Machine Learning

Looks good so far…



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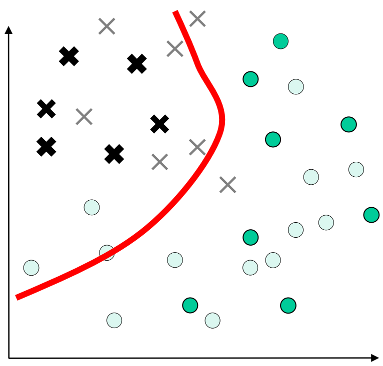
Over-fitting

An **Important** Concept in Machine Learning

Looks good so far…

Oh no! Mistakes!
What happened?



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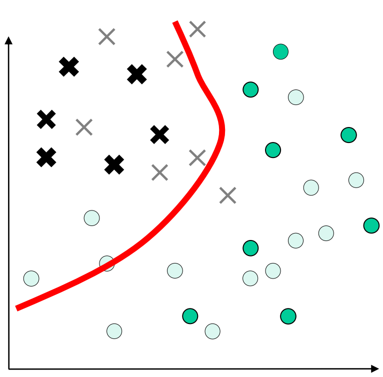
Over-fitting

An **Important** Concept in Machine Learning

Looks good so far…

Oh no! Mistakes!
What happened?



We didn't have all the data.

We can never assume that we do.

This is called "OVER-FITTING"
to the small dataset.

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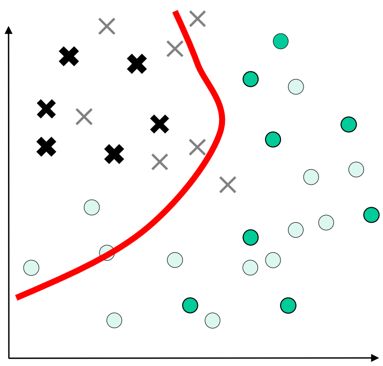
Over-fitting

- While an overly complex boundary may allow perfect classification of the training samples, it is unlikely to give good classification of novel patterns



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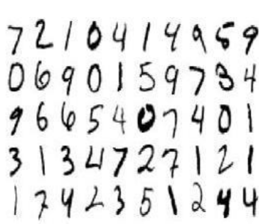
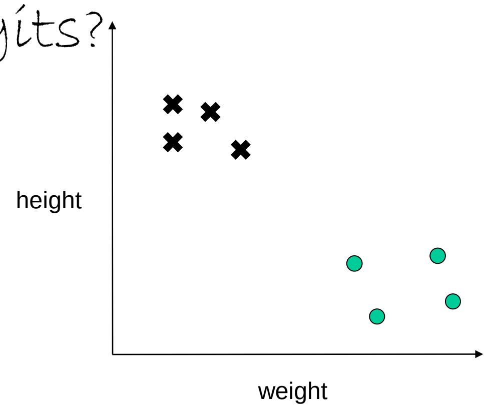
Our Problem

Now – how is this problem like recognising handwritten digits?





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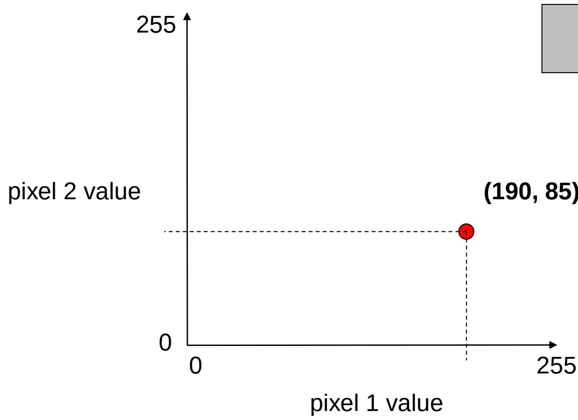
Recognising Handwritten Digits

Let's say the axes now represent **pixel values.**

A two-pixel image



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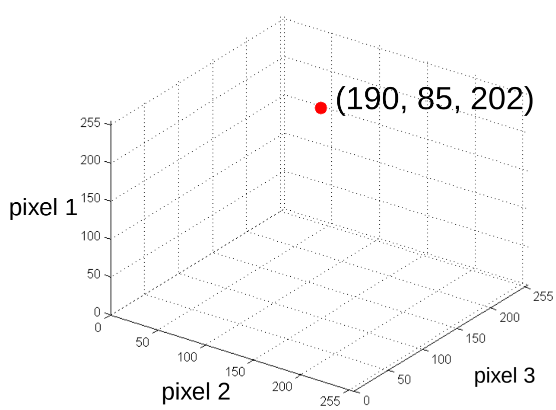
Recognising Handwritten Digits

Three dimensions…

A three-pixel image



This 3-pixel image is represented by a **SINGLE** point in a 3-D space.

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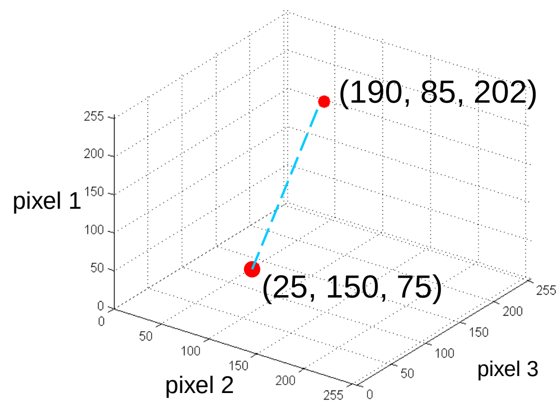
Recognising Handwritten Digits

Distances between images



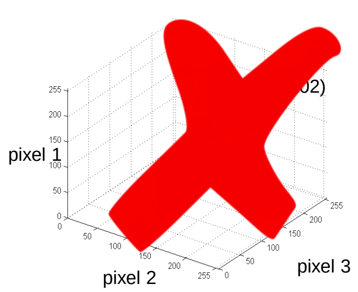
A three-pixel image

Another 3-pixel image

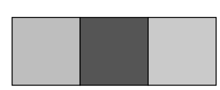
Straight line distance between them

Recognising Handwritten Digits

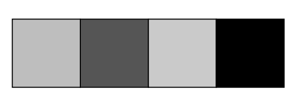
Four dimensions? Five? Six-dimensional spaces?



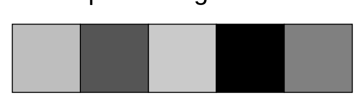
A three-pixel image



A four-pixel image.



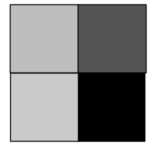
A five-pixel image



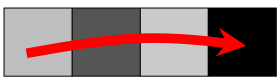
Recognising Handwritten Digits

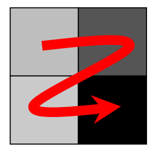
A four-pixel image.

A different four-pixel image.



(190, 85, 202, 10)
Same 4-dimensional vector!

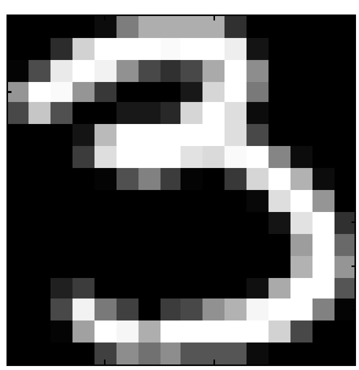




Assuming we read pixels in a systematic manner, we can now represent any image as a single point in a high dimensional space.

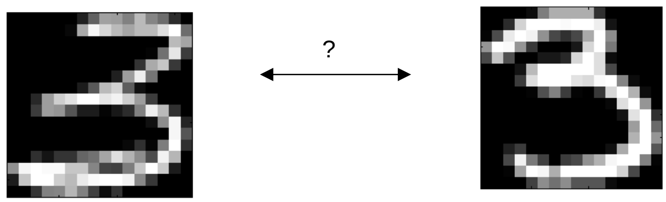
Recognising Handwritten Digits

16 x 16 image…. How many dimensions?

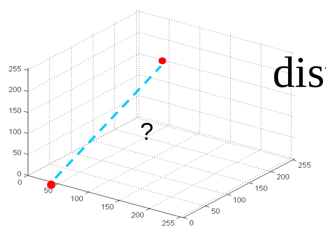


Recognising Handwritten Digits

Distances between digits….

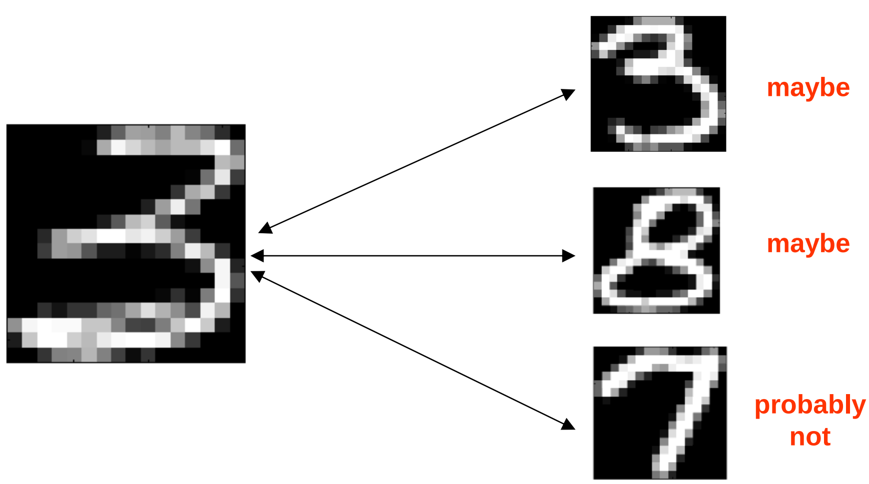


Straight-line distance in N-dimensional space?



$$\mathrm{distance}(x, x') = \sqrt{\sum_{i=1}^{i=256} (x_i - x'_i)^2}$$

Recognising Handwritten Digits

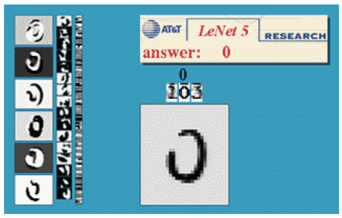


Which is **closest neighbour** in N-dimensions?

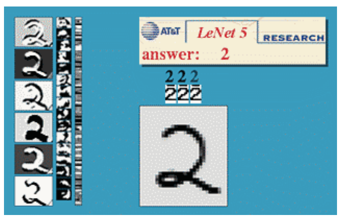
Recognising Handwritten Digits

AT&T Research Labs

The USPS ZIP code reader – **learnt** from examples.



FAST recognition



NOISE resistant, can **generalise** to future unseen patterns